

$$\begin{aligned}
 &= (CD \cdot A' + CD \cdot 1) + A \quad (\text{Postulate 2(b)}) \\
 &= (CD \cdot (A' + 1)) + A \quad (\text{Postulate 4(a)}) \\
 &= (CD \cdot 1) + A \quad (\text{Theorem 2(a)}) \\
 &= CD + A \quad (\text{Postulate 2(b)})
 \end{aligned}$$

(d) $(A+B)(A+B')(A+C)(A+C')$ to four literals

$$\begin{aligned}
 &= (A+B)(A+B')(A+C)(A+B') \quad (\text{Postulate 4(b)}) \\
 &= (A+B)(A+C)(A+B') \quad (\text{Postulate 5(b)}) \\
 &= (A+B)(A+C \cdot D)(A+B') \quad (\text{Postulate 4(b)}) \\
 &= (A+B)(A+C \cdot D \cdot 0)(A+B') \quad (\text{Postulate 3(b)}) \\
 &= (A+B)(A+C)(A+B') \quad (\text{Theorem 2(b)}) \\
 &= (A+B)(A+B') \quad (\text{Postulate 2(a)})
 \end{aligned}$$

2-7: Find the complement of the following Boolean functions and reduce them to a minimum number of literals.

(a) $(BC+A'D)(AB'+CD)'$ (Theorem 5(b))

Soln: $((BC+A'D)(AB'+CD))' = (BC+A'D)' + (AB'+CD)'$ (Theorem 5(a))

$$\begin{aligned}
 &= (BC)' + (A'D)' + (AB')' + (CD)'' \quad (\text{Theorem 5(b)}) \\
 &= (B'+C') + (A'+D') + (A+B) + (C+D) \quad (\text{Theorem 3}) \\
 &= (B'+C) + (A+D') + (A+B) + (C+D) \quad (\text{Postulate 4(a)}) \\
 &= (B'+C) + A + (B'+C) \cdot D' + (A+B) \cdot C + (A+B) \cdot D \quad (\text{Postulate 3(b)}) \\
 &= (A \cdot (B'+C) + D' \cdot (B'+C)) + (C' \cdot (A+B) + D \cdot (A+B)) \quad (\text{Postulate 3(b)}) \\
 &= (A \cdot B' + A \cdot C) + (D'B' + D'C) + (C'A + C'B) + (DA' + DB)
 \end{aligned}$$

(a) $(BC+A'D)(AB'+CD)'$

$$\begin{aligned}
 &= (BC+A'D)(AB') + (BC+A'D) \cdot (CD)' \quad (\text{Postulate 4(a)}) \\
 &= (AB') \cdot (BC+A'D) + (CD)' \cdot (BC+A'D) \quad (\text{Postulate 3(b)}) \\
 &= (AB') \cdot (BC') + (AB') \cdot (AD) + (CD)' \cdot (BC') + (CD)' \cdot (A'D) \quad (\text{Postulate 4(a)}) \\
 &= (A \cdot (B' \cdot (BC')) + (AB') \cdot (A'D)) + ((CD)' \cdot (BC') + C \cdot (D' \cdot (A'D))) \quad (\text{Theorem 4(b)}) \\
 &= (A \cdot (B' \cdot B) \cdot C') + (AB') \cdot (A'D) + ((CD)' \cdot (BC') + C \cdot (D' \cdot (A'D))) \quad (\text{Theorem 4(b)}) \\
 &= (A \cdot (B' \cdot B) \cdot C') + (BA') \cdot (A'D) + (BC') \cdot (CD)' + C \cdot (D' \cdot (DA')) \quad (\text{Postulate 3(b)}) \\
 &= (A \cdot (0 \cdot C')) + (BA') \cdot (A'D) + (BC') \cdot (CD)' + C \cdot (D' \cdot (DA')) \quad (\text{Postulate 5(b)})
 \end{aligned}$$